*A Mini Project*

**(Case Study Report)**

# Spatial and Predictive Analysis of Air Pollution in Indian Cities Using Machine Learning and Geospatial Techniques

***By***

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# 1. ABSTRACT

Air pollution is a major environmental and public health concern, particularly in rapidly urbanizing countries like India. This study aims to analyze the spatial and temporal patterns of air pollution across selected Indian cities from 2015 to 2023 using a combination of statistical, spatial, and machine learning techniques.

Exploratory data analysis revealed that AQI values are predominantly in the moderate range, with a right-skewed distribution indicating occasional extreme pollution events. Strong correlations were observed between AQI and particulate matter concentrations, especially PM2.5 and PM10. Spatial analysis using Moran's I, LISA, and Getis–Ord Gi* indicated no significant spatial autocorrelation or clustering, suggesting that AQI values are spatially independent at the city level. Spatial regression models further confirmed the absence of spatial dependence, while identifying PM2.5 as a highly significant predictor.

Predictive modeling using machine learning techniques demonstrated that ensemble models such as Random Forest and XGBoost outperform traditional regression approaches, achieving high accuracy in AQI prediction. Key predictors included PM2.5 and lagged AQI values, highlighting both environmental and temporal influences.

The study concludes that air pollution is primarily driven by local factors rather than spatial interactions between cities. It emphasizes the importance of city-level interventions and the use of data-driven approaches for effective air quality management.

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# 2. INTRODUCTION

## 2.1 Background

Air pollution is one of the most critical environmental and public health challenges faced globally in the 21st century. It refers to the presence of harmful substances in the atmosphere, including particulate matter, gases, and biological molecules, which can adversely affect human health, ecosystems, and climate. Among various pollutants, fine particulate matter (PM2.5), with a diameter of less than 2.5 micrometers, is particularly dangerous as it can penetrate deep into the lungs and bloodstream, leading to severe respiratory and cardiovascular diseases.

To measure and communicate the severity of air pollution, the Air Quality Index (AQI) is widely used. AQI is a standardized indicator that converts complex air quality data into a single numerical value, making it easier for the public and policymakers to understand pollution levels. It categorizes air quality into different levels such as "Good," "Moderate," "Poor," and "Hazardous," each associated with specific health implications. Thus, AQI serves as an essential tool for monitoring environmental quality and issuing health advisories.

In the context of India, air pollution has emerged as a major concern due to rapid urbanization, industrialization, vehicular emissions, and population growth. Several Indian cities frequently rank among the most polluted in the world, with AQI levels often exceeding safe limits prescribed by health organizations. Seasonal factors, such as winter inversion and crop residue burning, further aggravate pollution levels in certain regions. The variability in pollution levels across cities and over time makes it essential to conduct a detailed analysis that captures both spatial and temporal dimensions.

## 2.2 Problem Statement

Air pollution in India does not occur uniformly; instead, it exhibits significant variation across different geographical regions and time periods. Some cities consistently experience high pollution levels due to industrial activities and dense traffic, while others may show seasonal spikes influenced by meteorological conditions. This heterogeneity makes it difficult to draw generalized conclusions using traditional analytical approaches.

Moreover, pollution levels in one city may be influenced by neighboring regions due to the movement of air masses and regional environmental factors. This indicates the presence of spatial dependence, where nearby locations exhibit similar pollution characteristics. Ignoring such spatial relationships can lead to incomplete or misleading interpretations.

In addition, air pollution is inherently dynamic, changing over time due to human activities, policy interventions, and environmental conditions. Therefore, it is crucial to analyze pollution data not only across space but also across time. A combined spatio-temporal approach is necessary to fully understand the patterns, dependencies, and evolution of air pollution.

Despite the availability of large volumes of air quality data, there is a need for comprehensive analysis that integrates exploratory data analysis, spatial statistics, and predictive modeling. Such an approach can provide deeper insights into pollution patterns and enable better forecasting and decision-making.

## 2.3 Air Quality Index (AQI)

The Air Quality Index (AQI) is a standardized indicator used to communicate the quality of air and its potential impact on human health. It converts complex air pollution data into a single numerical value, making it easier for the general public to understand the level of air pollution in a specific area. The AQI typically ranges from 0 to 500, where higher values indicate higher levels of air pollution and greater associated health risks.

The AQI is calculated based on the concentrations of major air pollutants, including ground-level ozone, particulate matter (PM2.5 and PM10), carbon monoxide, sulfur dioxide, and nitrogen dioxide. These pollutants are monitored regularly, and their measured concentrations are transformed into index values. The overall AQI for a location is determined by the highest value among these pollutants, often referred to as the "dominant pollutant."

To enhance public understanding, the AQI is divided into six categories, each associated with a specific color, range, and level of health concern. For instance, AQI values between 0–50 indicate "Good" air quality with minimal health risk, while values above 300 are classified as "Hazardous," representing serious health threats to the entire population. As AQI levels increase, the severity of health effects also increases, particularly for sensitive groups such as children, the elderly, and individuals with respiratory or cardiovascular conditions.

The AQI serves as an important tool for environmental monitoring and public awareness. By simplifying complex environmental data into an easy-to-understand format, it enables individuals, communities, and policymakers to make informed decisions regarding outdoor activities and pollution control measures. It also plays a crucial role in issuing health advisories and implementing strategies to mitigate the adverse effects of air pollution.

### 2.3.1 What is PM2.5 (Fine Particulate Matter)

PM2.5 refers to fine particulate matter with a diameter of **2.5 micrometers or smaller**, which is approximately 30 times smaller than the width of a human hair. Due to their extremely small size, these particles can remain suspended in the air for long periods and can be easily inhaled into the human respiratory system.

PM2.5 particles originate from various sources, including vehicle emissions, industrial activities, construction dust, burning of fossil fuels, and biomass burning. Natural sources such as wildfires and dust storms can also contribute to PM2.5 levels. These particles may consist of a mixture of substances, including organic chemicals, metals, soil, and dust.

The significance of PM2.5 lies in its severe impact on human health. Unlike larger particles, PM2.5 can penetrate deep into the lungs and even enter the bloodstream, leading to respiratory and cardiovascular diseases. Prolonged exposure to high levels of PM2.5 is associated with increased risks of asthma, lung cancer, heart attacks, and premature mortality. Sensitive groups such as children, the elderly, and individuals with pre-existing health conditions are particularly vulnerable.

PM2.5 concentration is typically measured in micrograms per cubic meter (µg/m³) and is a key component in determining the Air Quality Index (AQI). Higher concentrations of PM2.5 generally correspond to poorer air quality and greater health risks. Monitoring PM2.5 levels is therefore essential for assessing environmental quality, issuing health advisories, and formulating pollution control policies.

In the context of this study, PM2.5 serves as a primary explanatory variable influencing AQI and plays a crucial role in both spatial analysis and predictive modeling of air pollution.

### 2.3.2 AQI Categories

The AQI value is then classified into categories to indicate health risk levels:

| **AQI Range** | **Category** | **Health Impact** |
|---|---|---|
| 0–50 | Good | Minimal impact |
| 51–100 | Moderate | Acceptable |
| 101–200 | Unhealthy for sensitive groups | Mild health effects |
| 201–300 | Poor | Breathing discomfort |
| 301–400 | Very Poor | Serious health effects |
| 401–500 | Hazardous | Severe health risk |

This means that higher AQI values indicate worse air quality. AQI simplifies multiple pollutant measurements into a single, easy-to-understand indicator. It helps in issuing health advisories and policy decisions. In this project, AQI is used as the primary dependent variable, representing overall air pollution levels. Since AQI is strongly influenced by pollutants like PM2.5, it serves as a key variable for both statistical analysis and predictive modeling.

### 2.3.3 Spatial Data, Areal Data, and Geospatial Data

Understanding air pollution requires not only statistical analysis but also a spatial perspective, as environmental phenomena are inherently linked to geographic location. In this context, **spatial data** plays a crucial role in analyzing and interpreting patterns of air quality across different regions.

**i) Spatial Data**

Spatial data refers to any data that is associated with a specific location on the Earth's surface. It typically includes two components:

- **Geometric information** (such as coordinates or boundaries)
- **Attribute information** (such as AQI, pollutant levels, or temperature)

Spatial data enables the study of how variables change across space and helps in identifying geographic patterns, dependencies, and relationships. In the context of air pollution, spatial data allows researchers to analyze how AQI varies across cities and regions.

**ii) Areal Data**

Areal data is a type of spatial data where observations are aggregated over defined geographic regions, such as cities, districts, or states. Instead of representing individual points, areal data represents values over areas.

In this study, AQI values aggregated at the **city level** represent areal data. Each city is treated as a spatial unit, and the average AQI or pollutant levels within that unit are analyzed.

**Importance:**

- Useful for policy-level analysis
- Enables comparison across administrative regions
- Simplifies complex spatial data into manageable units

**iii) Geospatial Data**

Geospatial data is a broader term that encompasses spatial data along with tools and technologies used to collect, store, analyze, and visualize geographic information. It integrates spatial coordinates with real-world attributes and is commonly used in Geographic Information Systems (GIS).

Geospatial data can be categorized into:

- **Vector Data:** Points (cities), lines (roads), polygons (regions)
- **Raster Data:** Grid-based data such as satellite imagery or pollution surfaces

In this study, geospatial data includes:

- City coordinates (latitude and longitude)
- Pollution indicators (AQI, PM2.5, etc.)
- Derived spatial features used for mapping and analysis

# 3. OBJECTIVE OF THE STUDY

The primary aim of this study is to perform a comprehensive spatio-temporal analysis of air pollution across major Indian cities using statistical and machine learning techniques.

The specific objectives are as follows:-

**i) To analyze pollution trends over time**

- Examine how AQI and PM2.5 levels change across different time periods.
- Identify seasonal patterns and long-term trends in air pollution.

**ii) To explore relationships between key variables**

- Study the association between PM2.5 concentration and AQI.
- Understand how different factors contribute to variations in air quality.

**iii) To detect spatial patterns and clustering**

- Apply spatial statistical techniques such as Moran's I to measure spatial autocorrelation.
- Identify pollution hotspots and cold spots using hotspot analysis methods.

**iv) To model spatial dependence in pollution data**

- Use spatial regression techniques to capture the influence of neighboring regions on pollution levels.
- Evaluate whether incorporating spatial effects improves model performance.

**v) To build predictive models for air quality**

- Develop regression and machine learning models such as Linear Regression, Random Forest, and XGBoost.
- Compare model performance using appropriate evaluation metrics.

**vi) To forecast future pollution levels**

- Use time series models to predict future AQI trends.
- Provide insights into potential future scenarios of air quality.

**vii) To support decision-making and policy formulation**

- Generate insights that can help in environmental planning and pollution control strategies.
- Highlight regions requiring urgent attention and intervention.

# 4. LITERATURE REVIEW

Spatial analysis and air pollution modelling have been extensively studied using statistical and geospatial techniques. Several foundational works and recent advancements provide the theoretical and methodological basis for analysing spatial dependence, clustering, and predictive modelling in environmental data.

One of the earliest and most significant contributions to spatial statistics is the work by **Moran (1950)**, who introduced the concept of spatial autocorrelation through Moran's I statistic. This measure evaluates whether similar values occur near each other in space, thereby identifying patterns of clustering or dispersion. Moran's I has since become a fundamental tool in spatial data analysis, widely applied in environmental studies to detect spatial dependence in variables such as air pollution.

Building on the concept of spatial dependence, **Getis and Ord** developed distance-based statistics for analyzing spatial association. Their work, *"The Analysis of Spatial Association by Use of Distance Statistics,"* introduced methods such as the Getis-Ord Gi* statistic, which is used to identify local clusters or "hotspots." This approach is particularly useful in identifying regions with significantly high or low pollution levels, making it highly relevant for environmental monitoring and policy planning.

In the domain of spatial regression, the work by **Colin M. Beale et al.** on *regression analysis of spatial data* highlights the importance of accounting for spatial autocorrelation in regression models. Traditional regression techniques often assume independence among observations; however, spatial data frequently violate this assumption. Spatial regression models, such as spatial lag and spatial error models, address this issue by incorporating spatial relationships into the modeling process, leading to more accurate and reliable results.

Recent advancements in spatial data science have been comprehensively documented in modern textbooks and practical guides. The book *"Spatial Statistics for Data Science: Theory and Practice with R"* provides a detailed overview of spatial statistical methods, including autocorrelation, interpolation, and spatial modeling, along with practical implementation. Similarly, *"Spatial Data Science with Applications in R"* by Edzer Pebesma and Roger Bivand emphasizes applied spatial analysis techniques and their use in real-world datasets.

Furthermore, the book *"Geographic Data Science with Python"* by Sergio J. Rey, Dani Arribas-Bel, and Levi J. Wolf highlights the integration of spatial analysis with modern programming tools. It demonstrates how Python-based libraries can be used for geospatial visualization, spatial econometrics, and machine learning, making it highly relevant for contemporary data science applications.

Overall, the existing literature underscores the importance of combining spatial statistics, regression techniques, and computational tools to analyze geographically distributed data. These studies provide a strong foundation for the present work, which integrates exploratory data analysis, spatial autocorrelation, hotspot detection, and predictive modeling to analyze air pollution patterns across Indian cities.

# 5. DATA COLLECTION AND DESCRIPTION

## 5.1 Data Collection

The data used in this study was collected from publicly available and reliable sources that provide air quality measurements across major Indian cities. The primary focus was on key air pollution indicators such as Air Quality Index (AQI) and particulate matter concentration (PM2.5), which are widely recognized as critical measures of air pollution.

The dataset was obtained from platforms such as the Central Pollution Control Board (CPCB) and open-source repositories (e.g., Kaggle/OpenAQ), which provide historical air quality data for multiple cities in India. The data spans several years (e.g., 2015–2023), allowing for both spatial and temporal analysis of pollution trends.

To enable spatial analysis, geographical coordinates (latitude and longitude) corresponding to each city were incorporated into the dataset. These coordinates were either directly available in the dataset or obtained through geocoding methods using publicly available geographic information.

The collected data includes daily observations of AQI and PM2.5 levels for multiple cities. Since raw data often contains missing or inconsistent values, preprocessing steps such as handling missing values, filtering relevant variables, and converting date formats were performed to ensure data quality and consistency.

Additionally, for analytical convenience and to reduce noise, the data was aggregated into monthly averages. This helped in capturing broader pollution trends while minimizing short-term fluctuations.

## 5.2 Data Description

### 5.2.1 Pollution Dataset (*pollution*)

This dataset contains daily air quality measurements for multiple Indian cities over several years. It includes the Air Quality Index (AQI) along with concentrations of major air pollutants, making it the primary dataset for analysis.

**i. Dimensions:**
32,870 rows and 10 columns

**ii. Key Variables and Data Types:**

| Variable | Data Type | Description |
|---|---|---|
| city | Object (string) | Name of the city |
| date | Object (converted to datetime) | Date of observation |
| pm25 | Float64 | Concentration of PM2.5 (fine particulate matter) |

| | | |
|---|---|---|
| pm10 | Float64 | Concentration of PM10 |
| no2 | Float64 | Nitrogen dioxide concentration |
| so2 | Float64 | Sulfur dioxide concentration |
| co | Float64 | Carbon monoxide concentration |
| o3 | Float64 | Ozone concentration |
| aqi | Int64 | Air Quality Index |
| aqi_category | Object (string) | AQI classification (e.g., Good, Moderate, Unhealthy) |

**iii. Time Range:**
1st January 2015 to 31st December 2023
This dataset forms the core of the study and is used for exploratory analysis, spatial modeling, and predictive modeling.

### 5.2.2. Weather Dataset (*weather*)

This dataset provides meteorological conditions along with certain air quality indicators for various locations. It is used to incorporate environmental factors that may influence air pollution levels.

**i. Dimensions:**
572 rows and 42 columns

**ii. Key Variables and Data Types (relevant to this study):**

| **Variable** | **Data Type** | **Description** |
|---|---|---|
| location_name | Object (string) | Name of the location |
| latitude, longitude | Float64 | Geographical coordinates |
| last_updated | Object (converted to datetime) | Timestamp of the observation |
| temperature_celsius | Float64 | Temperature in degrees Celsius |
| wind_kph | Float64 | Wind speed (km/h) |
| humidity | Int64 | Relative humidity (%) |
| pressure_mb | Float64 | Atmospheric pressure (millibars) |
| precip_mm | Float64 | Precipitation (mm) |
| air_quality_PM2.5 | Float64 | PM2.5 concentration |

| | | |
|---|---|---|
| air_quality_PM10 | Float64 | PM10 concentration |
| air_quality_Ozone | Float64 | Ozone concentration |
| air_quality_Carbon_Monoxide | Float64 | CO concentration |
| air_quality_Nitrogen_dioxide | Float64 | $NO_2$ concentration |
| air_quality_Sulphur_dioxide | Float64 | $SO_2$ concentration |

**ii. Time Range:**
Data available only for a single date: 29th August 2023

**iii. Remarks:**
Due to its limitation of representing only one day, this dataset was not directly suitable for temporal analysis. Therefore, simulated weather data was generated based on this dataset to extend its usability across the full time span of the pollution dataset.

### 5.2.3. Coordinates Dataset (*coords*)

This dataset provides geographical coordinates and population information for a large number of cities and locations. It was primarily used to ensure accurate spatial mapping of pollution data.

**i. Dimensions:**
557,957 rows and 4 columns

**ii. Key Variables and Data Types:**

| **Variable** | **Data Type** | **Description** |
|---|---|---|
| name | Object (string) | Name of the city/location |
| latitude, longitude | Float64 | Geographical coordinates |
| population | Int64 | Population of the location |

**iii. Remarks:**
This dataset served as a reference for obtaining consistent and accurate latitude and longitude values for cities in the pollution dataset. Only relevant entries corresponding to the cities in the pollution dataset were utilized during the merging process.

## 5.3 Data Preparation and Preprocessing

**i. Data Loading and Initial Inspection**

The analysis began with the loading of three primary datasets: pollution data (air quality indicators), weather data (meteorological conditions), and city coordinates data (geographical information). These datasets were imported into the working environment using the pandas library, specifically the pd.read_csv() function, which converts raw CSV files into structured DataFrames.

Initial inspection of the datasets was performed to understand their structure, variable types, and presence of missing values. This step ensured familiarity with the data and helped identify potential inconsistencies that required preprocessing.

Libraries Used: pandas
Contribution: This step established the foundational datasets required for subsequent analysis and modeling.

**ii. Standardization of City Names and Integration of Geographical Coordinates**

A key challenge in integrating multiple datasets was the inconsistency in city naming conventions across the pollution and coordinates datasets. To address this, city names were standardized by converting all entries to lowercase and removing leading and trailing whitespace using string operations such as .str.lower() and .str.strip().

Following standardization, the coordinates dataset was merged with the pollution dataset using a left join on the city column. This ensured that all pollution records were retained while incorporating the corresponding latitude and longitude information. The resulting dataset, referred to as *pollution_with_coords*, contained both pollution measurements and their associated spatial attributes.

Libraries Used: pandas (string operations and DataFrame merging using pd.merge())
Contribution: This step enriched the dataset with spatial information, making it suitable for geospatial analysis and mapping.

**iii. Date Handling and Weather Data Integration**

To ensure temporal consistency, the date columns in both the pollution and weather datasets were converted into datetime format using pd.to_datetime(). This enabled accurate time-based operations such as sorting, filtering, and merging.

However, a limitation was observed in the weather dataset, which contained information for only a single date (e.g., 2023-08-29), whereas the pollution dataset spanned multiple years (2015–2023). To overcome this limitation, weather variables such as temperature, wind speed, humidity, pressure, and precipitation were simulated for all cities and dates present in the pollution dataset.

The simulation was performed by taking the available weather data as a baseline and introducing small random variations using a normal distribution (numpy.random.normal()), thereby generating realistic but synthetic daily weather values.

The simulated weather dataset was then merged with the pollution and coordinate data to create a comprehensive dataset (*pollution_with_simulated_weather*), containing pollution, spatial, and meteorological variables.

Libraries Used: pandas (date handling and merging), numpy (random number generation)

Contribution: This step enabled the inclusion of weather-related variables in the analysis, enhancing the ability to study environmental influences on air pollution despite limited real-world data availability.

### iv. Final Data Preparation for Modeling

To prepare the dataset for predictive modeling, additional features were engineered to capture temporal dependencies. These included:

- Lag variables (e.g., previous day AQI) using .shift()
- Rolling statistics (e.g., moving averages) using .rolling()
- Grouped operations using .groupby() for city-wise calculations

These operations introduced missing values at the beginning of the series. Such rows were removed using the .dropna() function to ensure that the final dataset contained complete observations.

The resulting dataset, referred to as *pollution_modeling_data*, was clean, consistent, and suitable for machine learning and statistical modeling.

Libraries Used: pandas (feature engineering and missing value handling)
Contribution: This step ensured data quality and created meaningful features, improving the performance and reliability of predictive models.

GeoDataFrame head:

| | city | latitude | longitude | year | month | aqi | pm25 | pm10 | no2 | so2 | co | o3 | simulated_temperature_celsius | simulated_wind_kph | simulated_humidity | s |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | ahmedabad | 23.02579 | 72.58727 | 2015 | 1 | 91.516129 | 84.979723 | 134.613581 | 40.953590 | 12.764292 | 0.777701 | 50.592002 | 29.881288 | 11.937431 | 58.013144 | |
| **1** | ahmedabad | 23.02579 | 72.58727 | 2015 | 2 | 87.107143 | 79.476451 | 131.053654 | 38.725969 | 11.515238 | 0.814806 | 46.281887 | 30.681739 | 12.810510 | 57.140332 | |
| **2** | ahmedabad | 23.02579 | 72.58727 | 2015 | 3 | 84.806452 | 74.362866 | 129.920191 | 41.095574 | 10.424440 | 0.776220 | 50.232294 | 29.801851 | 12.705055 | 60.898283 | |
| **3** | ahmedabad | 23.02579 | 72.58727 | 2015 | 4 | 103.900000 | 100.230590 | 151.944586 | 41.072665 | 12.612991 | 0.875261 | 45.850634 | 27.858865 | 13.044750 | 59.444850 | |
| **4** | ahmedabad | 23.02579 | 72.58727 | 2015 | 5 | 87.903226 | 79.882882 | 132.143893 | 40.642157 | 11.803615 | 0.840628 | 47.652168 | 29.897628 | 11.165014 | 61.572206 | |

| simulated_pressure_mb | simulated_precip_mm | geometry |
|---|---|---|
| 1017.207753 | 0.091540 | POINT (72.58727 23.02579) |
| 1000.457358 | 0.065479 | POINT (72.58727 23.02579) |
| 1011.579483 | 0.089412 | POINT (72.58727 23.02579) |
| 1001.388764 | 0.072806 | POINT (72.58727 23.02579) |
| 995.521818 | 0.095714 | POINT (72.58727 23.02579) |

# 6. METHODOLOGY

This study employs a combination of spatial statistics, regression modeling, and geostatistical techniques to analyze the spatial and temporal dynamics of air pollution. The methodology is grounded in the principle that **geographical data exhibit spatial dependence**, meaning that observations located near each other tend to have similar values.

## 6.1 Concept of Spatial Dependence

A fundamental concept in spatial analysis is **spatial autocorrelation**, which is based on *Tobler's First Law of Geography*:

"Everything is related to everything else, but near things are more related than distant things."

This implies that air pollution levels in nearby cities are likely to be more similar than those in distant cities. Therefore, traditional statistical methods that assume independence of observations may not be appropriate, necessitating the use of spatial techniques.

## 6.2 Spatial Autocorrelation (Moran's I)

Spatial autocorrelation is used to quantify the degree of similarity between observations across space. In this study, **Global Moran's I** is used to measure the overall spatial dependence of AQI values.

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$$

Where:

- $n$= number of observations
- $x_i, x_j$= AQI values at locations $i$and $j$
- $\bar{x}$= mean AQI
- $w_{ij}$= spatial weights indicating proximity between locations

**i. Theoretical Insight:**

Moran's I summarizes spatial patterns similarly to how the mean summarizes a distribution. It captures the relationship between a variable and its spatial lag, i.e., the average value of neighboring observations .

**ii. Interpretation:**

$\boldsymbol{H_0}$: $\boldsymbol{I = E[I]}$ $(\boldsymbol{no\ spatial\ autocorrelation})$

$H_1 : I \neq E[I]$ (*spatial autocorrelation*)

- $I > 0$: Positive spatial autocorrelation (clustering)
- $I < 0$: Negative spatial autocorrelation (dispersion)
- $I \approx 0$: Random spatial distribution

**iii. Application:**

This study uses Moran's I to test whether high or low AQI values are geographically clustered across Indian cities.

## 6.3 Spatial Lag and Spatial Regression

**i. Spatial Lag Concept**

A key concept in spatial modeling is the spatial lag, defined as:

$$Y_{sl} = WY$$

This represents the weighted average of neighboring values and captures local spatial influence .

**ii. Need for Spatial Regression**

In conventional regression models (e.g., OLS), observations are assumed to be independent. However, spatial data often violate this assumption, as residuals may exhibit spatial clustering, indicating unmodeled spatial structure .

**iii. Spatial Lag Model (SLM)**

To account for spatial dependence, the Spatial Lag Model is used:

$$y = \rho W y + X\beta + \epsilon$$

Where:

- $y$= AQI (dependent variable)
- $X$= explanatory variables (PM2.5, weather factors)
- $Wy$= spatial lag of AQI
- $\rho$= spatial autoregressive coefficient
- $\epsilon$= error term

**iv. Interpretation:**

- A significant $\rho$indicates spillover effects, meaning pollution in one city is influenced by nearby cities

- This captures regional pollution dynamics

**v. Application:**

Spatial regression is used to model AQI while accounting for geographic dependency, improving prediction accuracy and reducing bias.

## 6.4 Geostatistical Interpolation (Kriging)

Kriging is a geostatistical method used to estimate values at unobserved locations based on spatial correlation.

**i. Ordinary Kriging**

$$Z^*(s_0) = \sum_{i=1}^{n} \lambda_i Z(s_i)$$

Where:

- $Z^*(s_0)$= predicted AQI at location $s_0$
- $Z(s_i)$= observed AQI at known locations
- $\lambda_i$= weights based on spatial correlation

**ii. Variogram (Core of Kriging)**

$$\gamma(h) = \frac{1}{2} E[(Z(s) - Z(s+h))^2]$$

- Measures how similarity decreases with distance
- Forms the basis for assigning weights in Kriging

**iii. Theoretical Insight:**

Kriging assumes that spatial variation is continuous and smooth, and nearby points provide more information than distant ones. It is closely related to spatial smoothing and Gaussian process models .

**iv. Application:**

In this study, Kriging is used to:

- Estimate AQI in regions without monitoring stations
- Generate continuous pollution maps

## 6.5 Integration with Predictive Modelling

To complement spatial analysis, machine learning models are applied:

- Linear Regression (baseline)
- Random Forest (non-linear modeling)
- XGBoost (high-performance boosting)
- ARIMA (time series forecasting)

These models incorporate both spatial features (coordinates) and temporal features (lags, rolling means).

## 6.6 Model Evaluation

Model performance is evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Coefficient of Determination ($R^2$)

These metrics assess the predictive accuracy and reliability of the models.

## 6.7 Local Indicators of Spatial Association (LISA)

While Global Moran's I provides an overall measure of spatial autocorrelation, it does not identify *where* clustering occurs. To address this limitation, Local Indicators of Spatial Association (LISA) are used to detect local spatial patterns.

**i. Concept**

LISA decomposes global spatial autocorrelation into contributions for each observation, enabling the identification of local clusters and spatial outliers.

**Local Moran's I**

$$I_i = (x_i - \bar{x}) \sum_j w_{ij} (x_j - \bar{x})$$

Where:

- $I_i$= Local Moran's I for location $i$
- $x_i$= AQI value at location $i$
- $\bar{x}$= mean AQI
- $w_{ij}$= spatial weight between locations $i$and $j$

**ii. Cluster Types Identified**

LISA classifies observations into the following categories:

- **High–High (HH):** High AQI surrounded by high AQI → pollution hotspots
- **Low–Low (LL):** Low AQI surrounded by low AQI → cleaner regions
- **High–Low (HL):** High AQI surrounded by low AQI → spatial outliers
- **Low–High (LH):** Low AQI surrounded by high AQI → spatial anomalies

**iv. Application in Study**

LISA is used to identify **localized clusters of air pollution**, allowing for targeted analysis of regions with significantly high or low AQI levels. This provides deeper insights beyond global spatial patterns.

## 6.8 Hotspot Analysis (Getis–Ord Gi*)

To further identify statistically significant clusters of high and low values, **Getis–Ord Gi*** statistics are applied.

**i. Concept**

The Getis–Ord Gi* statistic measures the degree of concentration of high or low values in a neighborhood, identifying **hotspots and cold spots**.

$$G_i^* = \frac{\sum_j w_{ij}\, x_j - \bar{X} \sum_j w_{ij}}{S\sqrt{\frac{n \sum_j w_{ij}^2 - (\sum_j w_{ij})^2}{n-1}}}$$

Where:

- $x_j$= AQI at location $j$
- $w_{ij}$= spatial weight
- $\bar{X}$= mean AQI
- $S$= standard deviation
- $n$= number of observations

**ii. Interpretation**

- High positive values → **Hotspots** (high AQI clusters)
- High negative values → **Cold spots** (low AQI clusters)
- Values near zero → no significant clustering

**iii. Application in Study**

This method is used to identify statistically significant pollution hotspots, helping in pinpointing regions with consistently high air pollution levels.

## 6.9 Spatial Visualization using Folium

To effectively communicate spatial patterns, interactive maps were generated using the Folium library.

**i. Visualization Approach**

- Each city is represented as a **point on the map** using latitude and longitude
- AQI values are:
  - Color-coded (e.g., green for low AQI, red for high AQI)
  - Scaled by marker size to reflect intensity
- Pop-ups display detailed information such as city name and AQI

**ii. LISA Cluster Visualization**

The results of LISA analysis were also visualized on interactive maps:

- Different colors were used to represent cluster types:
  - Red → High–High clusters
  - Blue → Low–Low clusters
  - Other colors → spatial outliers
- This allowed for intuitive identification of **regional pollution patterns**

**iii. Importance**

Interactive spatial visualization provides:

- Better understanding of geographic patterns
- User-friendly exploration of data
- Enhanced communication of results to non-technical audiences

## 6.10 Predictive Modelling Approach

To forecast AQI levels, both statistical and machine learning models are applied:

- **Linear Regression**: Baseline model
- **Random Forest**: Captures non-linear relationships
- **XGBoost**: Gradient boosting for high accuracy
- **ARIMA**: Time series forecasting

## 6.11 Model Evaluation Metrics

The performance of predictive models is evaluated using the following metrics:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Coefficient of Determination ($R^2$)

These metrics assess the accuracy and reliability of the models in predicting AQI values.

# 7. IMPLEMENTATION AND PROCEDURE

## 7.1. Prepare Cross-Sectional Data for Spatial Analysis (city_gdf)

To perform spatial analysis, the dataset was transformed into a cross-sectional format by aggregating data at the city level. The GeoDataFrame (*gdf*), containing monthly observations, was grouped by city, and the mean values of AQI, pollutant concentrations, simulated weather variables, and geographical coordinates were computed.

The resulting dataset (*city_gdf*) represents each city with its average characteristics over the study period.

- **Libraries Used:** geopandas for creating the GeoDataFrame and pandas for grouping and aggregation.
- **Contribution:** This step created a spatially consistent dataset suitable for autocorrelation, clustering, and regression analysis.

## 7.2. Spatial Weights Matrix (w)

A spatial weights matrix was constructed using the K-Nearest Neighbors (KNN) approach with $k = 4$. Each city was assigned four nearest neighbors based on geographical distance, and the matrix was row-standardized.

- **Libraries Used:** libpysal.weights.KNN for creating the spatial weights matrix.
- **Contribution:** This matrix quantified spatial relationships between cities and served as a key input for spatial statistical methods.

## 7.3 Global Spatial Autocorrelation (Moran's I)

Global Moran’s I was computed using AQI values and the spatial weights matrix to assess the presence of spatial autocorrelation. Statistical significance was evaluated using permutation-based testing.

- **Libraries Used:** esda.moran.Moran.
- **Contribution:** This provided an overall measure of whether AQI values exhibit clustering or randomness across cities.

## 7.4 Local Spatial Autocorrelation (LISA - Moran_Local)

Local Moran’s I statistics were calculated to identify location-specific spatial patterns. Each city was classified into clusters such as High–High, Low–Low, High–Low, Low–High, or Not Significant based on its local statistic and significance level.

- **Libraries Used:** esda.moran.Moran_Local.
- **Contribution:** This enabled identification of localized pollution clusters and spatial outliers.

### 7.5 Hotspot Analysis (Getis-Ord G - G_Local)

The Getis–Ord Gi* statistic was computed to detect statistically significant hotspots and cold spots of AQI. Classification was based on Z-scores and associated p-values.

- **Libraries Used:** esda.getisord.G_Local.
- **Contribution:** This provided an alternative perspective on spatial clustering, focusing specifically on regions of high and low concentration.

## 7.6 Spatial Regression (ML_Lag and ML_Error)

- **Purpose:** To explicitly model the relationship between AQI and influencing factors (like PM2.5 and simulated temperature) while accounting for spatial dependence. We used two types of spatial regression models:
  - **Spatial Lag Model (ML_Lag):** Assumes that the AQI in a city is directly influenced by the AQI in neighboring cities. It includes a spatially lagged dependent variable (W_AQI).
  - **Spatial Error Model (ML_Error):** Assumes that spatial dependence is present in the error terms, meaning unobserved factors affecting AQI are spatially correlated.
- Both models were fitted using city_gdf['aqi'] as the dependent variable and simulated_temperature_celsius and pm25 as independent variables. A constant term was added to the independent variables.
- **Libraries Used:** spreg.ML_Lag, spreg.ML_Error, and statsmodels.api for adding the constant.
- **Contribution:** These models helped determine if spatial spillover effects were significant and how they affected the relationship between AQI and other variables. The comparison of model fit statistics (AIC, BIC, R-squared) helped in understanding which model better captured the spatial processes.

## 7.7 Kriging for Spatial Interpolation

- To estimate AQI values at unobserved locations and create a continuous surface map of AQI across the region, based on the discrete city measurements.
- Ordinary Kriging was performed using the longitude, latitude, and average AQI from city_gdf. A spherical variogram model was chosen, and the interpolation was executed over a dense grid covering the study area.
- **Libraries Used:** pykrige.ok.OrdinaryKriging.
- **Contribution:** Kriging provided a visual representation of AQI distribution beyond the specific city points, suggesting patterns and potential hot/cold areas where direct measurements were not available.

## 7.8 Spatial Visualization with Folium

- To create interactive maps that visually represent the spatial patterns and results of the analyses, making them easy to interpret.
- Folium was used to generate three interactive maps:
    - **Average AQI Levels:** Displays each city with a circle marker scaled and colored by its average AQI, with detailed tooltips.
    - **LISA Clusters:** Colors each city based on its LISA classification (High-High, Low-Low, High-Low Outlier, Low-High Outlier, Not Significant).
    - **Hotspot Clusters:** Colors each city based on its Getis-Ord G classification (Hot Spot, Cold Spot, Not Significant).
- **Libraries Used:** folium for creating interactive maps.
- **Contribution:** These visualizations are critical for communicating the spatial findings effectively, allowing for intuitive exploration of AQI distribution and clustering.

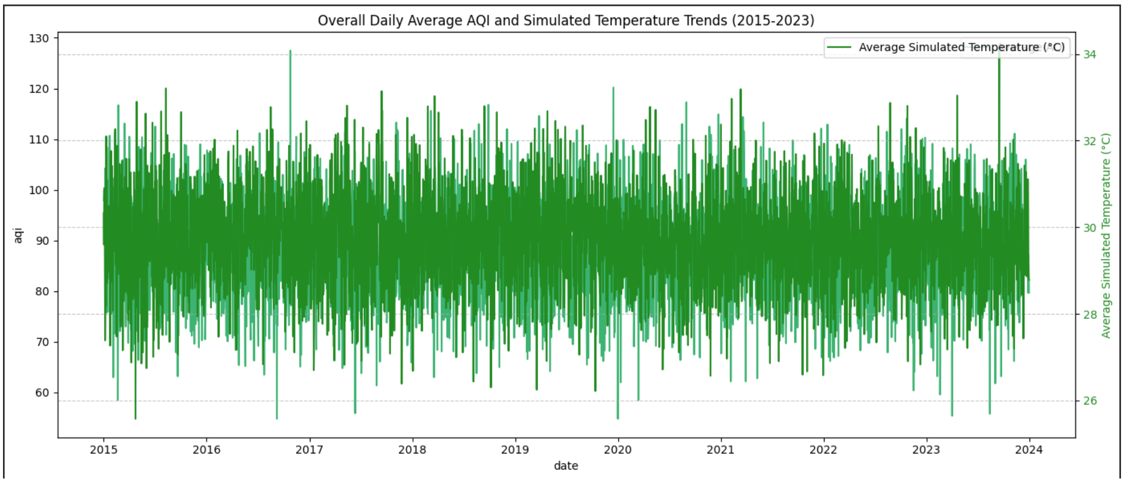
# 8. RESULTS AND DISCUSSION

## 8.1 Exploratory Data Analysis (EDA) and Descriptive Statistics

### 8.1.1 Temporal Trends in Air Pollution and Temperature

The time series analysis of daily average AQI across all cities from 2015 to 2023 reveals noticeable temporal variability, with alternating periods of relatively higher and lower pollution levels. The overall mean AQI was approximately 88, which falls within the *Moderate* air quality category, indicating that, on average, air quality remains acceptable but may pose risks for sensitive groups.

The simulated temperature series exhibited expected seasonal patterns, with higher values during summer months and lower values during winter. However, the Pearson correlation coefficient between AQI and simulated temperature was found to be 0.0036, indicating a negligible linear relationship between these variables at the aggregated level.

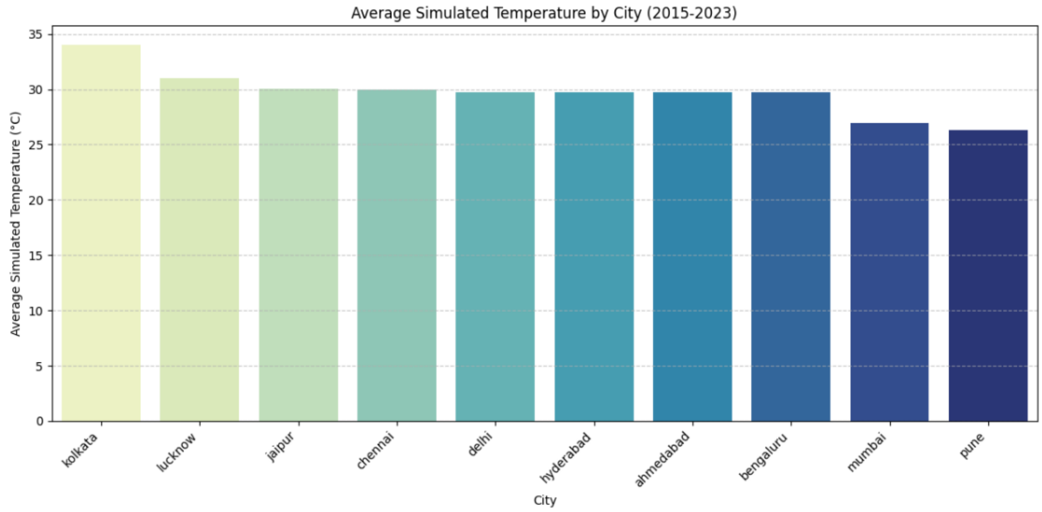



**Interpretation:**
This suggests that temperature alone is not a strong determinant of AQI when considered across all cities and time periods. Other factors such as emissions and particulate matter likely play a more dominant role.

### 8.1.2 Spatial Variation Across Cities

The comparison of average AQI values across cities revealed moderate spatial variation. Cities such as Ahmedabad, Hyderabad, and Pune exhibited relatively higher average AQI levels, whereas Mumbai and Chennai showed comparatively lower values. However, the overall range of AQI across cities remained narrow, clustering around the global mean.

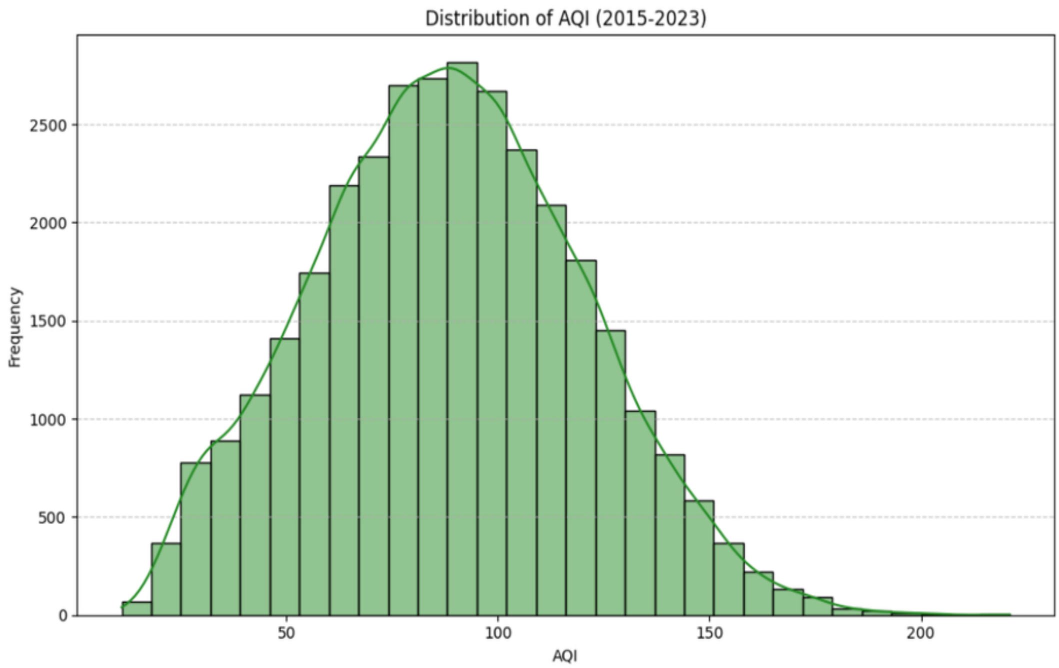
Similarly, simulated temperature varied geographically, with cities like Kolkata and Lucknow showing higher average temperatures, while coastal cities such as Mumbai and Pune exhibited relatively lower values.



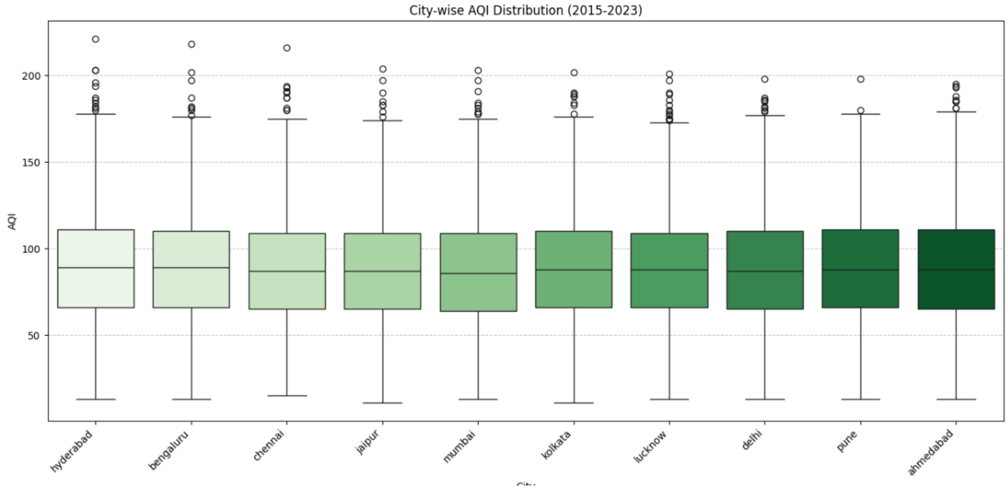

**Interpretation:**
These findings indicate that while there are city-specific differences, pollution levels are broadly similar across major Indian cities, suggesting the presence of shared contributing factors such as urbanization and traffic emissions.

### 8.1.3 Distributional Characteristics of AQI

The histogram of AQI values revealed a **right-skewed distribution**, indicating that lower to moderate AQI levels are more frequent, with fewer instances of extremely high pollution levels. The peak of the distribution was concentrated in the lower range of the *Moderate* category.




City-wise box plots further highlighted the presence of variability and **outliers**, with several cities exhibiting occasional spikes in AQI.

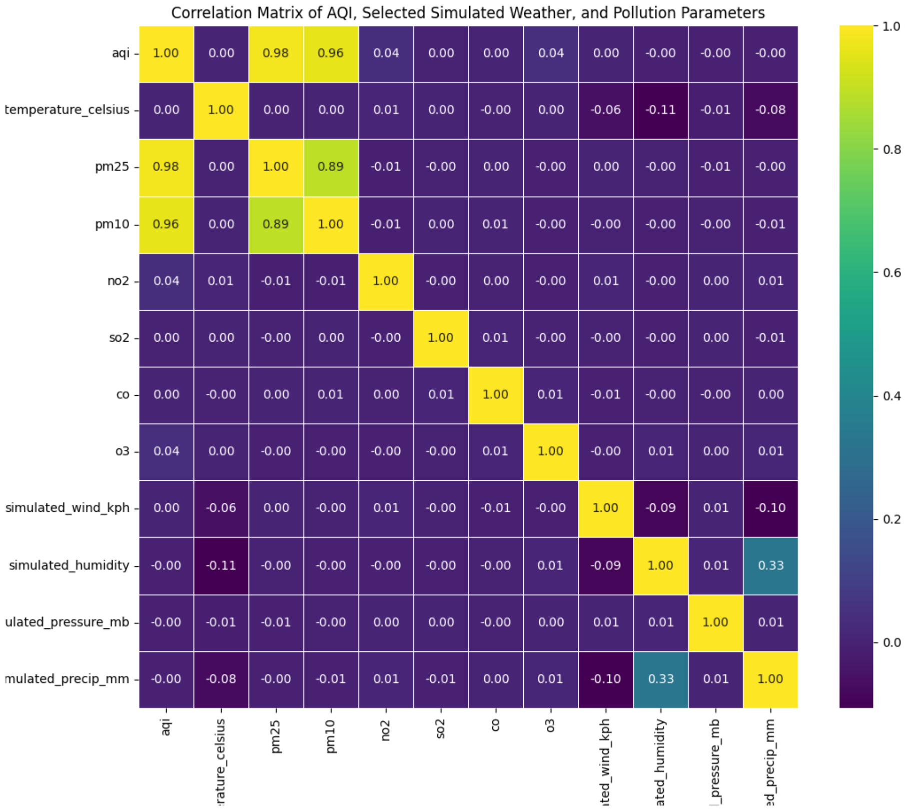



**Interpretation:**
The skewness indicates that while extreme pollution events are less frequent, they do occur and may significantly impact overall air quality. These spikes could be associated with episodic events such as festivals, industrial emissions, or seasonal factors.

### 8.1.4 Correlation Analysis

The correlation analysis provided key insights into the relationships between AQI and other variables:

- **Strong Positive Correlations:**
  PM2.5 and PM10 showed extremely high positive correlations with AQI (approximately **0.99**), confirming their dominant role in determining air quality. Other pollutants such as $NO_2$, $SO_2$, CO, and $O_3$ also exhibited moderate to strong positive correlations.
- **Weather Variables:**
  Simulated weather variables demonstrated weak or negligible correlations with AQI. Temperature showed near-zero correlation, while humidity had a slight positive association. Wind speed and pressure showed weak negative correlations.




**Interpretation:**
These results confirm that **particulate matter (PM2.5 and PM10) is the primary driver of AQI**, whereas weather variables, at least in this simulated and aggregated context, do not exhibit strong direct linear relationships.

### 8.1.5 Descriptive Statistics

The summary statistics of the aggregated dataset indicated that:

- The mean AQI was approximately **88.1**, with relatively low variability (standard deviation ≈ 0.6 in monthly aggregated data)
- Pollutant concentrations showed expected ranges and dispersion across cities
- City-wise statistics revealed slight differences in pollution levels but overall consistency in trends

**Interpretation:**
The relatively low variability in aggregated data suggests stability in average pollution levels over time, although short-term fluctuations and extreme values still occur.

City-wise Statistics (PM2.5, AQI):

| | pm25 | | | | aqi | | | |
|---|---|---|---|---|---|---|---|---|
| | **mean** | **std** | **min** | **max** | **mean** | **std** | **min** | **max** |
| **city** | | | | | | | | |
| **ahmedabad** | 81.180027 | 7.284470 | 63.611798 | 100.654925 | 88.489181 | 6.029024 | 74.166667 | 105.387097 |
| **bengaluru** | 81.059854 | 6.754974 | 56.805269 | 98.467681 | 88.366279 | 5.537566 | 67.354839 | 102.451613 |
| **chennai** | 80.461865 | 6.964895 | 64.978915 | 101.222219 | 87.883154 | 5.750367 | 75.233333 | 104.483871 |
| **delhi** | 80.763571 | 6.797320 | 61.914697 | 94.807580 | 88.258449 | 5.669466 | 72.107143 | 101.766667 |
| **hyderabad** | 81.615058 | 7.656077 | 61.368284 | 103.715942 | 88.874822 | 6.337576 | 73.033333 | 105.806452 |
| **jaipur** | 80.290611 | 7.197608 | 55.702845 | 98.826431 | 87.620076 | 5.857667 | 65.967742 | 100.806452 |
| **kolkata** | 80.932544 | 6.243815 | 64.848848 | 100.116269 | 88.290632 | 5.112969 | 77.064516 | 103.419355 |
| **lucknow** | 80.347307 | 6.637377 | 65.115667 | 98.025064 | 87.845136 | 5.243887 | 76.333333 | 100.451613 |
| **mumbai** | 79.323273 | 7.093356 | 59.112797 | 102.867002 | 86.853230 | 5.748021 | 73.266667 | 106.193548 |
| **pune** | 81.427733 | 7.201009 | 59.313093 | 97.237005 | 88.595203 | 5.820165 | 70.064516 | 101.400000 |

### 8.1.6 Skewness and Kurtosis

Most pollutant variables, including AQI, PM2.5, and PM10, exhibited **positive skewness**, confirming the right-tailed distributions observed earlier.

Kurtosis values indicated varying degrees of peakedness:

- Some variables were **leptokurtic**, suggesting the presence of extreme values
- Others were closer to normal distribution

**Interpretation:**
The presence of skewness and high kurtosis highlights the occurrence of **extreme pollution events**, which are critical from a public health perspective and should not be overlooked in analysis.

## 8.2 Spatial Analysis Results and Discussion

This section presents the findings from spatial statistical techniques applied to analyze the geographic distribution and dependence of AQI across selected Indian cities.

### 8.2.1 Global Spatial Autocorrelation (Moran's I)

The Global Moran's I statistic for AQI was computed as **0.0036**, with a corresponding p-value of **0.4079**.

**Interpretation:**
The Moran's I value is very close to zero and statistically insignificant ($p > 0.05$), indicating the absence of global spatial autocorrelation. This suggests that AQI values across cities are **randomly distributed in space**, with no significant clustering of high or low values.

**Discussion:**
This result implies that, at the aggregated city level, air pollution does not exhibit strong regional patterns. Unlike many environmental variables that show spatial dependence, AQI in this dataset appears to be driven more by **local city-specific factors** rather than broader geographic influences.

### 8.2.3 Local Spatial Autocorrelation (LISA)

The Local Indicators of Spatial Association (LISA) analysis was conducted to identify localized spatial clustering and outliers in AQI values across cities.

The LISA cluster classification revealed the following:

- Not Significant: 9 cities
- Low–High Outlier: 1 city (Mumbai)

Specifically, Mumbai was identified as a Low–High outlier with a statistically significant p-value ($p = 0.0070$), while all other cities did not exhibit statistically significant local spatial autocorrelation.

**Interpretation**

A Low–High outlier indicates that:

- The city itself has a relatively low AQI
- It is surrounded by neighboring cities with relatively higher AQI values

In this case, Mumbai has lower AQI compared to its neighboring cities, making it spatially distinct.

### 8.2.4 Hotspot Analysis (Getis–Ord Gi*)

The Getis–Ord Gi* analysis was performed to identify statistically significant hotspots and cold spots of AQI.

- **Hotspots:** None
- **Cold spots:** None
- **Not Significant:** 10 cities

**Interpretation:**
No statistically significant hotspots or cold spots were detected at the 95% confidence level.

**Discussion:**
This result further confirms the lack of spatial concentration of high or low AQI values. Unlike typical urban pollution studies where hotspots are observed in industrial or densely populated regions, this dataset does not exhibit such patterns, possibly due to **limited sample size (10 cities)** or aggregation effects.

### 8.2.5 Spatial Regression Analysis

Spatial regression models were employed to examine the relationship between AQI and its determinants while accounting for spatial dependence.

**i. Key Findings:**

- **PM2.5 (pm25):**
  A highly significant predictor of AQI (**$p < 0.001$**) in both models. Higher PM2.5 concentrations strongly increase AQI levels.
- **Simulated Temperature:**
  Not statistically significant in either model, indicating minimal influence on AQI.

**ii. Spatial Lag Model (ML_Lag)**

- Spatial coefficient ($\rho$): **-0.1934**
- p-value: **0.2152** (not significant)

**iii. Interpretation:**
The spatial lag effect is not statistically significant, suggesting that AQI in one city is not influenced by AQI in neighboring cities.

**iv. Spatial Error Model (ML_Error)**

- Spatial error coefficient ($\lambda$): **-0.3076**

- p-value: **0.5740** (not significant)

**v. Interpretation:**
There is no evidence of spatial dependence in the error terms, indicating that **unobserved factors affecting AQI are not spatially correlated**.

**vi. Model Performance**

- Pseudo $R^2$:
  - ML_Lag: **0.9895**
  - ML_Error: **0.9879**
- Model Comparison:
  - ML_Error showed slightly better fit (lower AIC and SIC)

**Discussion:**
The very high $R^2$ values are primarily driven by the strong explanatory power of PM2.5. However, the lack of significance of spatial parameters ($\rho$ and $\lambda$) indicates that spatial effects do not play a meaningful role in explaining AQI variation in this dataset.

**Conclusion from Regression:**
A non-spatial regression model may be sufficient, as spatial dependence is not statistically supported.

### 8.2.6 Kriging Interpolation

Kriging interpolation was used to generate a continuous spatial surface of AQI across the study region.

**i. Observations:**

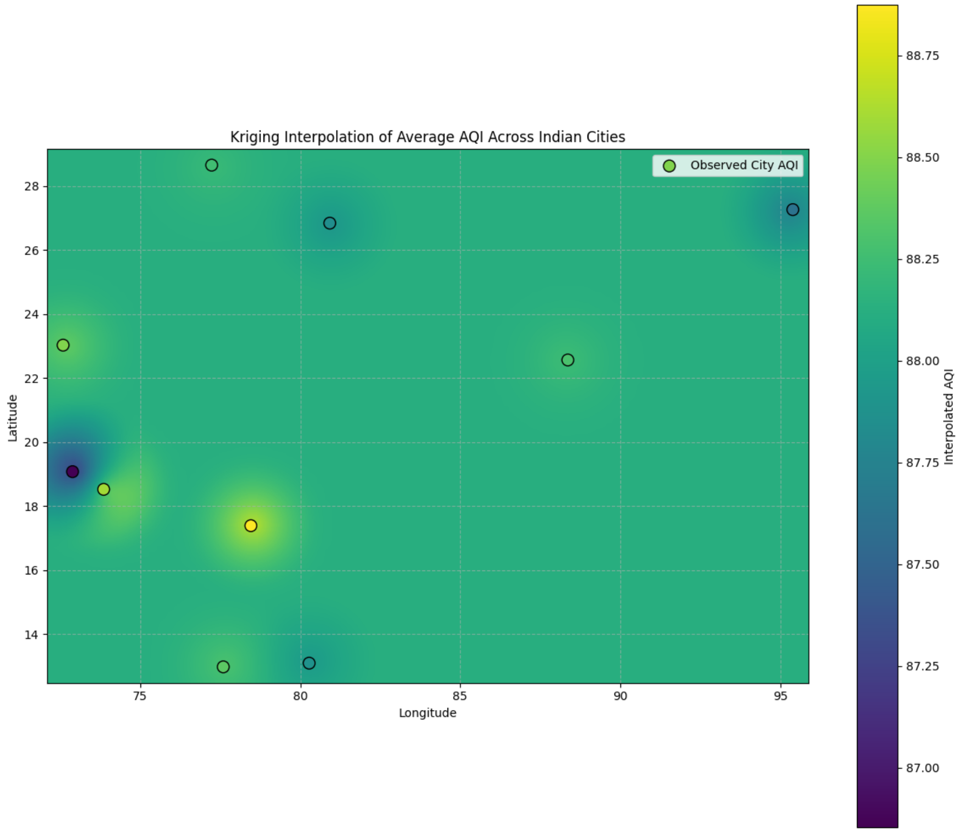
- The interpolated AQI values are **fairly uniform across the region**, mostly ranging between **~87 to 89**.
- Slightly **higher AQI zones (yellow-green regions)** are observed around central parts of the map (around longitude ~78–80), indicating relatively higher pollution levels in those areas.
- **Lower AQI regions (blue-green shades)** are visible near the western coastal side (around longitude ~72–74), suggesting comparatively better air quality.
- The variation across space is **very smooth**, with no sharp gradients or abrupt changes.

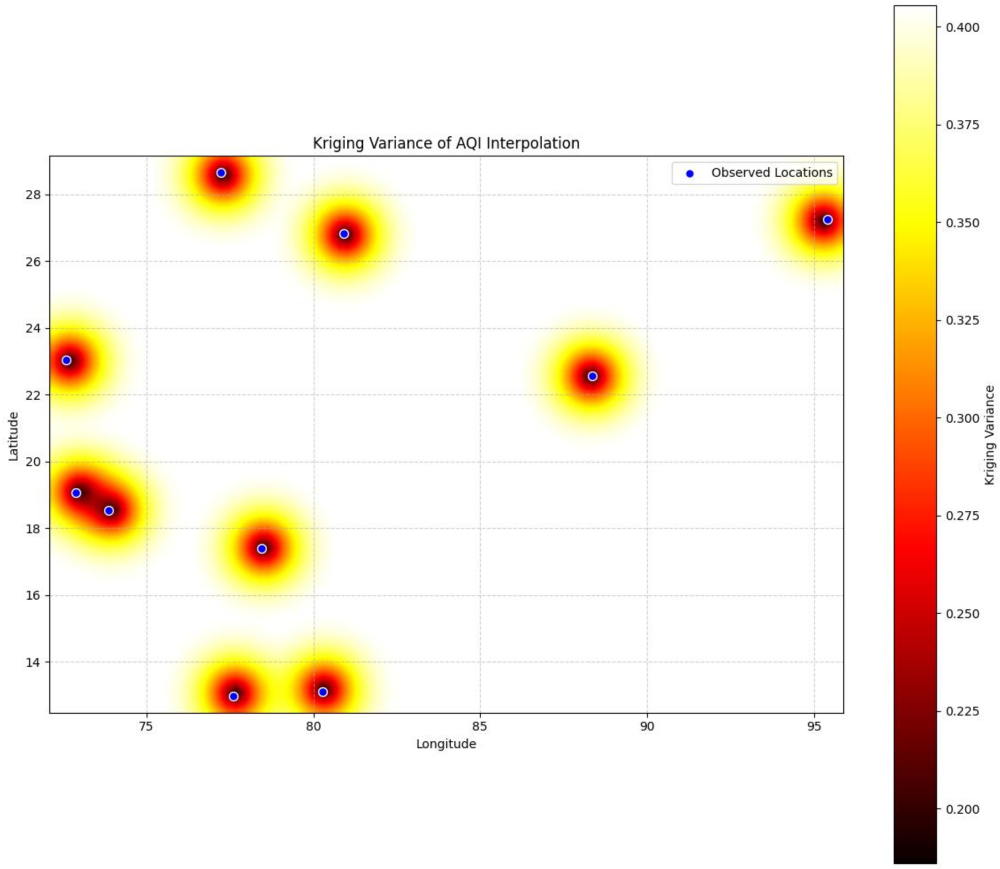
ii. **Interpretation:**

The smooth and nearly uniform AQI surface suggests that:

- There is **low spatial variability** in AQI across the selected cities
- Pollution levels are **relatively similar across locations**

- No strong regional pollution clusters are evident



iii. **Kriging Variance (Uncertainty Map)**

The Kriging variance map represents the uncertainty associated with the interpolated AQI values.

**Observations:**

- Lower variance (darker regions) is observed near the actual city locations (blue points), indicating higher confidence in predictions near observed data.
- Higher variance (yellow/red regions) appears in areas farther away from observed cities, indicating greater uncertainty.
- The uncertainty increases gradually as the distance from known data points increases.

**Interpretation:**

- Kriging predictions are most reliable near observed data points
- Predictions in regions far from cities are less reliable and more uncertain
- The spatial distribution of variance highlights the importance of data density

### 8.2.7 OVERALL SPATIAL INSIGHTS

The spatial analysis leads to the following key conclusions:

- No significant **global or local spatial autocorrelation** was detected
- No **hotspots or cold spots** of AQI were identified
- **PM2.5 is the dominant factor** influencing AQI
- Spatial regression models indicate **no significant spatial spillover effects**
- AQI variations are primarily driven by **local factors rather than neighboring influences**
- Overall, the findings suggest that air pollution patterns in this dataset are **spatially independent at the city level**. Unlike typical environmental phenomena that exhibit clustering, AQI appears to be influenced predominantly by **city-specific emission sources and conditions**, rather than geographic proximity.

## 8.3 Predictive Modelling Results and Discussion

To evaluate the predictive capability of different models, three approaches—Linear Regression, Random Forest, and XGBoost—were implemented and compared using standard performance metrics.

### 8.3.1 Model Performance Comparison

| Model | MAE | RMSE | $R^2$ Score |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Random Forest | 8.07 | 10.37 | 0.9706 |
| XGBoost | 8.23 | 10.52 | 0.9698 |
| Linear Regression | 14.66 | 18.74 | 0.9016 |

### 8.3.2 Interpretation of Results

The results indicate that **ensemble machine learning models significantly outperform the traditional linear regression model**.

- **Random Forest** achieved the best performance, with the lowest error values and highest $R^2$ score (~0.971), demonstrating excellent predictive accuracy.
- **XGBoost** also performed comparably well, indicating strong capability in capturing complex relationships in the data.
- **Linear Regression**, while providing a reasonable baseline, showed comparatively higher error values and lower explanatory power.

**Discussion:**
The superior performance of Random Forest and XGBoost suggests that the relationship between AQI and its predictors is **non-linear and complex**, which cannot be fully captured by simple linear models.

### 8.3.3 Feature Importance and Key Predictors

Analysis of feature importance revealed that:

- **PM2.5 (pm25)** is the most influential predictor of AQI
- **Lagged AQI (aqi_lag1)** plays a significant role, indicating temporal persistence
- **Rolling mean of AQI (aqi_roll_mean)** further enhances predictive performance

**Interpretation:**
These findings confirm that AQI is strongly influenced by:

- Current pollutant levels (especially PM2.5)
- Past pollution levels (temporal dependency)

This highlights the importance of both **environmental and temporal factors** in air quality prediction.

# 9. CONCLUSION BASED ON ANALYSIS

Based on the comprehensive spatial and machine learning analyses of AQI data across selected Indian cities from 2015 to 2023, the following key conclusions can be drawn:

## 9.1 Spatial Analysis Insights

- No statistically significant global spatial autocorrelation was observed (Moran's I ≈ 0)
- LISA analysis did not identified 9 not significant local clusters or and 1 spatial outliers
- Hotspot analysis revealed no statistically significant hotspots or cold spots
- Spatial regression models showed:
    - PM2.5 as a highly significant predictor
    - No significant spatial lag (ρ) or spatial error (λ) effects

**Conclusion:**
AQI variations are primarily driven by local factors within cities, rather than spatial spillover effects or inter-city dependencies.

## 9.2 Spatial Interpolation Insight

Kriging interpolation provided a continuous visualization of AQI distribution; however:

- Results are sensitive to the limited number of cities (n = 10)
- Interpretation should be approached with caution

**Conclusion:**
The Kriging interpolation results indicate a relatively uniform spatial distribution of AQI with low variability across regions. However, the associated variance map highlights increased uncertainty in areas distant from observed locations, suggesting that the reliability of interpolation is constrained by the limited spatial coverage of the dataset..

## 9.3 Machine Learning Insights

- **Random Forest and XGBoost** demonstrated superior predictive performance
- **Linear Regression** was less effective in capturing complex patterns
- **PM2.5, lagged AQI, and rolling averages** were the most influential predictors

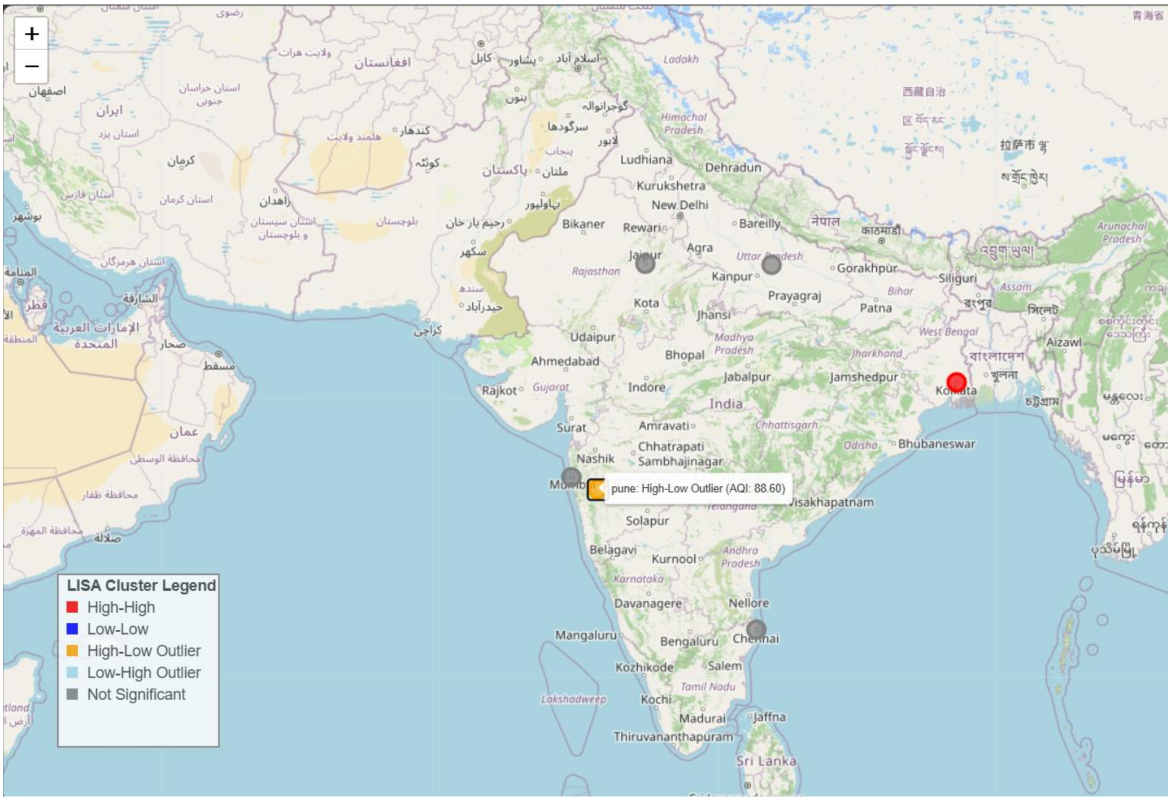
**Conclusion:**
AQI can be accurately predicted using machine learning models that capture **non-linear relationships and temporal dependencies**.

## 9.4 Overall Conclusion

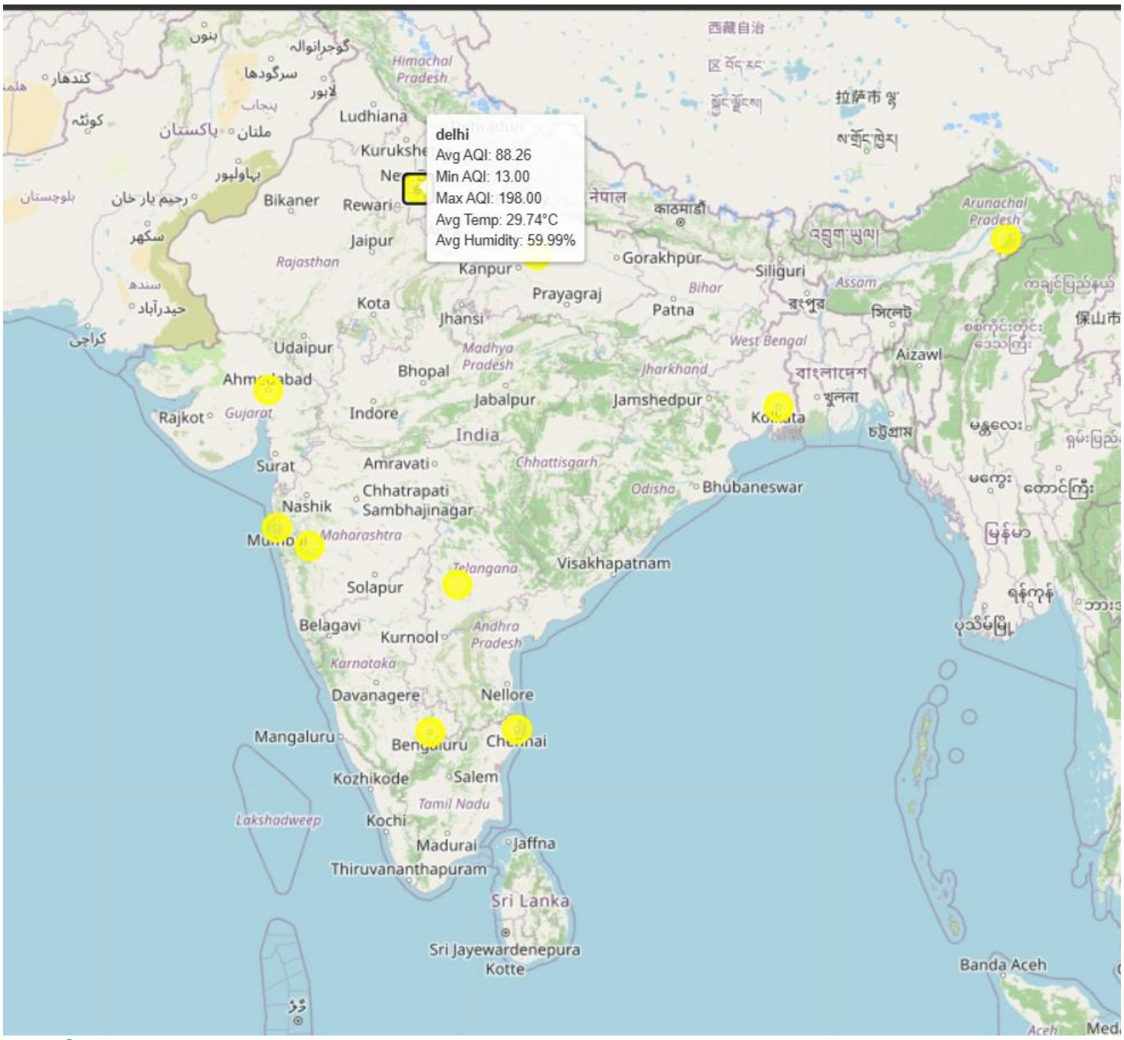
The study reveals that air pollution patterns across the selected Indian cities are largely spatially independent at the aggregated level. Unlike many environmental phenomena, AQI does not exhibit significant spatial clustering or spillover effects in this dataset.Instead, AQI is predominantly influenced by - Local pollutant concentrations (especially PM2.5),Temporal persistence of pollution levels.

## 9.5 Folium Maps

Folium maps are created by first initializing a base map using geographic coordinates and a zoom level. Spatial features such as markers, circles, or polygons are then added to represent data points. These features can be customized with colors, sizes, and interactive elements like tooltips or popups. Finally, the map can be displayed within the notebook environment or saved as an HTML file for external viewing. Following are some maps made in analysis.




lisa_clusters_map.html




average_aqi_map (2).html (Command Line)

# 10. SUMMARY AND CONCLUSION OF THE ANALYSIS

This study conducted a comprehensive analysis of air pollution across selected Indian cities from 2015 to 2023 by integrating **descriptive statistics, exploratory data analysis (EDA), spatial analysis, and predictive modeling techniques**.

The exploratory analysis revealed that AQI values predominantly fall within the *Moderate* category, with noticeable temporal fluctuations over the years. The distribution of AQI was found to be **right-skewed**, indicating that while most observations correspond to moderate pollution levels, occasional extreme pollution events do occur. Strong positive correlations were observed between AQI and particulate matter concentrations, particularly **PM2.5 and PM10**, confirming their dominant role in determining air quality. In contrast, simulated weather variables showed weak or negligible relationships with AQI.

Spatial analysis was performed to examine whether air pollution exhibits geographic dependence across cities. However, the results from **Global Moran's I**, **LISA**, and **Getis–Ord Gi*** indicated the absence of statistically significant spatial autocorrelation, clustering, or hotspot formation. This suggests that AQI values are **spatially independent** at the aggregated city level. Spatial regression models further confirmed this finding, as spatial parameters (lag and error components) were not statistically significant. Instead, **PM2.5 emerged as the most significant predictor of AQI**, reinforcing the importance of local pollution sources.

The Kriging interpolation results indicate a relatively uniform spatial distribution of AQI with low variability across regions. However, the associated variance map highlights increased uncertainty in areas distant from observed locations, suggesting that the reliability of interpolation is constrained by the limited spatial coverage of the dataset.

Finally, predictive modeling demonstrated that **machine learning models significantly outperform traditional regression approaches**. Random Forest and XGBoost achieved high predictive accuracy, capturing non-linear relationships and temporal dependencies effectively. Key predictors included PM2.5, lagged AQI, and rolling averages, indicating strong persistence in pollution patterns.

**CONCLUSION**

The findings of this study highlight that air pollution, as measured by AQI, is primarily influenced by **local factors rather than spatial interactions between cities**. Despite applying multiple spatial statistical techniques, no significant spatial clustering or dependence was observed.

The analysis consistently identifies **PM2.5 as the most critical determinant of AQI**, emphasizing its role in air quality degradation. Additionally, the presence of temporal dependence suggests that past pollution levels strongly influence current conditions.

While spatial methods did not reveal significant geographic patterns, machine learning models demonstrated excellent predictive capabilities, indicating that AQI dynamics are better captured through **data-driven approaches that account for non-linearity and temporal structure**.

Overall, the study concludes that effective air pollution management strategies should focus on **local emission control and pollutant reduction**, rather than relying on regional spillover assumptions.

# 11. FUTURE SCOPE

The study can be extended and improved in several ways:

I. **Inclusion of More Cities and Monitoring Stations**
Expanding the dataset to include a larger number of cities and finer spatial resolution would improve the robustness of spatial analysis and may reveal hidden clustering patterns.

II. **Use of Real-Time and High-Frequency Weather Data**
Incorporating actual weather data instead of simulated values could provide deeper insights into the interaction between meteorological conditions and air pollution.

III. **Incorporation of Additional Variables**
Factors such as traffic density, industrial activity, land use patterns, and population density can enhance the explanatory power of the models.

IV. **Advanced Spatio-Temporal Models**
Future studies can explore advanced models such as **Geographically Weighted Regression (GWR)** or **spatio-temporal deep learning models** to better capture dynamic dependencies.

V. **Policy Impact Analysis**
Evaluating the impact of government policies and interventions on AQI levels can provide actionable insights for environmental planning.

# 12. DATA SOURCES AND REFERENCES

## 12.1 DATA SOURCES

### I. Air Pollution Data

- Central Pollution Control Board (CPCB), Government of India
- Open-source platforms like Kaggle and other environmental data repositories

### II. Weather Data

- Online weather data services (e.g., Weather API datasets available on Kaggle)

### III. Geographic Coordinates Data

City-level geographic coordinates (latitude and longitude) and population data were obtained from:

- Global city datasets available on Kaggle
- Open geospatial data repositories

## 12.2 Tools and Libraries

The analysis was conducted using the following tools and libraries:

- Python (Programming Language)
- pandas, numpy (Data processing)
- geopandas (Spatial data handling)
- libpysal, esda, spreg (Spatial statistics)
- pykrige (Spatial interpolation)
- folium (Interactive mapping)
- scikit-learn, xgboost (Machine learning models)

## 12.3 REFERENCES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